

The background of the entire image is a dark blue field filled with a pattern of red dots. These dots are arranged in a way that they form a large, faint circular shape in the center, with the density of the dots being higher in the center and tapering off towards the edges. The dots are of varying sizes, creating a textured, digital effect.

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The Core Objective

Train a parametric model $f(\cdot; \theta)$ on a set of N available **source domains** $\mathcal{S} = \{S_1, S_2, \dots, S_N\}$ such that it generalizes robustly to K completely **unseen target domains** $\mathcal{T} = \{T_1, T_2, \dots, T_K\}$.

The Strict Constraint vs. Domain Adaptation

- ▶ **Domain Adaptation (DA)**: Has access to unlabeled target domain data during training.
- ▶ **Domain Generalization (DG)**: **Zero access** to target-domain data or target labels during the optimization cycle.

The ERM Trap

- ▶ Models minimize error by taking the **path of least resistance**.
- ▶ Exploits **superficial shortcuts** (textures, backgrounds).

Domain Shift Collapse

- ▶ **Photo** → **Sketch**: Context and style cues completely vanish.
- ▶ **Result**: Catastrophic accuracy drop.

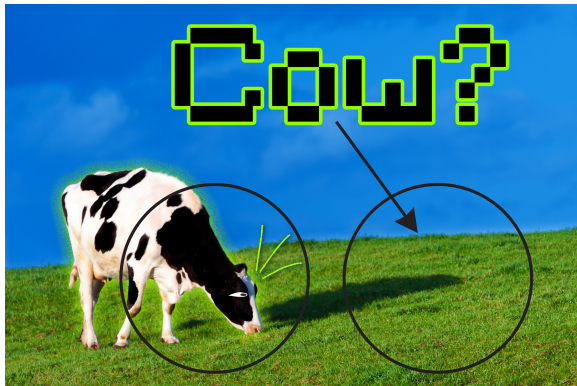


Fig: ERM relies on "Grass" (Shortcut) instead of the actual "Cow" geometry.

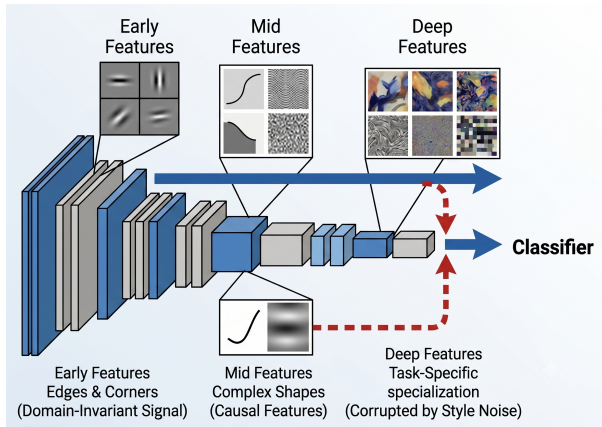
The Last-Layer Problem

- ▶ Standard CNNs use **deepest features** only.

The Invariant Signal

- ▶ **Early/Mid Layers:** Hold shapes and edges.

⇒ Corrupted by style and filters out raw structure details.

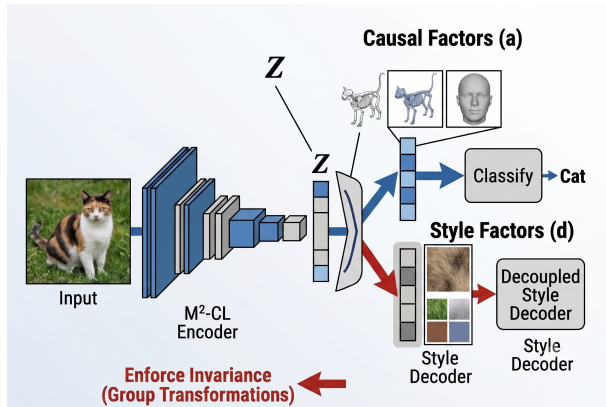


Latent Space Split: $X \approx (a, d)$

- ▶ **Causal (a):** Class-specific geometry (Invariant).
- ▶ **Style (d):** Domain-specific context and texture (Noise).

The DG Goal

Isolate **Causal** a from **Style** d to ensure the model only relies on invariant geometry for prediction.



Disentangled Group Actions

Let group $G = G_1 \times \cdots \times G_n$ act on world space $X = X_1 \times \cdots \times X_n$:

$$(g_1, \dots, g_n) \cdot (x_1, \dots, x_n) = (g_1 \cdot_1 x_1, \dots, g_n \cdot_n x_n)$$

$\implies$ Each sub-space X_i is strictly invariant to all subgroup actions G_j ($j \neq i$).

The M^2 Latent Hypothesis

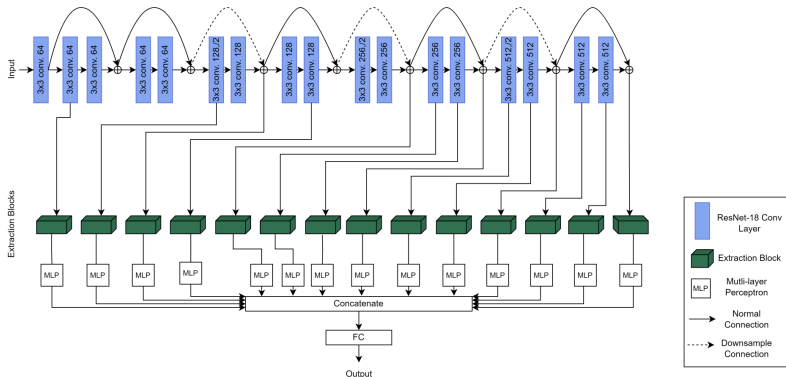
Instead of a single entangled final layer, we decompose intermediate hidden layers:

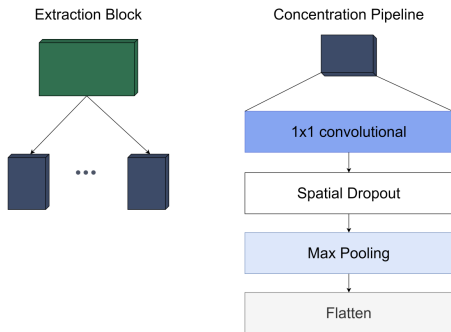
$$Z = Z_1 \times Z_2 \times \cdots \times Z_{L-1}$$

$\implies$ Extraction modules handle high-dimensionality and filter out redundant info.

Framework Concept

Leverages parallel multi-scale concentration blocks mapped across multiple intermediate layers of a standard backbone network.





Concentration pipeline layout.

Internal Sequence Mechanics

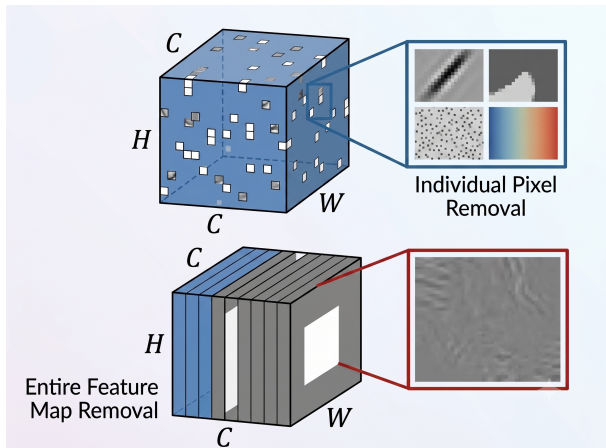
1. **1×1 Convolutional Layer:** Compresses high-dimensional channels c down to $\lfloor c/r \rfloor$ using a reduction factor r .
2. **Spatial Dropout Layer:** Mitigates pixel correlation leakage.
3. **Multi-Scale Max Pooling:** Captures spatial features before reduction.
4. **MLP Projection Vector:** Maps pooled features into vectors via MLPs.

Vanilla Dropout (CNN Failure)

- ▶ Drops **individual pixels** randomly.
- ▶ **Result:** Info leaks; regularization fails.

Spatial Dropout (Our Solution)

- ▶ Drops **entire feature maps** ($H \times W$).
- ▶ Forces network to find alternative cues.
- ▶ **Result:** Blocks local style shortcuts.



Spatial Scaling Allocations

- ▶ **Early Hooks:** Pooling sizes optimized for 8×8 , 4×4 , and 2×2 .
- ▶ **Late Hooks:** Deeper fields adapt to 7×7 and 3×3 dimensions.

Concatenation Protocol

Flattened multi-scale vectors are concatenated into a **hypercolumn vector** for final FC layer classification.

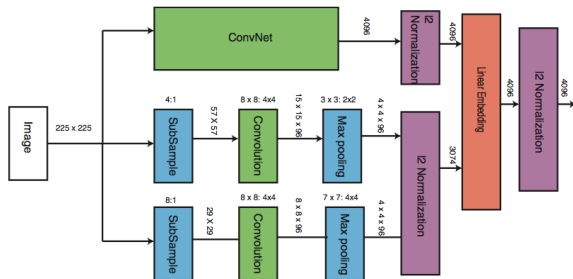


Fig: Multi-scale pooling pipelines merging into a unified Hypercolumn vector.

The Necessity of Geometric Constraints

Simply extracting intermediate representations does not ensure that hidden filters actively discard domain-specific style noise.

Target Geometric Alignment

- ▶ **Pull Same-Class Closely:** Force representations of identical semantic labels (e.g., a Dog in Photo and a Dog in Sketch) to align tightly in the latent space across domains.
- ▶ **Push Different-Classes Apart:** Move different semantic labels far apart, even if they share an identical background style.

Probability Alignment Measure

Let $u_i^{(l)}$ be the L2-normalized extraction vector ($\|u_i^{(l)}\| = 1$) of sample x_i at hidden layer l . The probability alignment measure of class c within a batch is defined as:

$$p^{(l)}(c) = \frac{\sum_{i=1}^{N_c} \sum_{j=i+1}^{N_c} \exp(u_i^{(l)T} u_j^{(l)} / \tau)}{\sum_{i=1}^{N_b} \sum_{a \neq i}^{N_b} \exp(u_i^{(l)T} u_a^{(l)} / \tau)}$$

Parameters Key

- ▶ N_c : Total count of elements belonging strictly to class c inside the batch.
- ▶ N_b : Complete batch sample size count.
- ▶ τ : Temperature parameter regulating distribution density mass.

Joint Structural Loss

Considering all C distinct classes, the total log-likelihood regularizer loss over M selected hidden layers is formulated as:

$$\mathcal{L}^{(l)} = - \sum_{c=1}^C \log p^{(l)}(c) \implies \mathcal{L} = \mathcal{L}_{CE} + \alpha \sum_{l=1}^M \mathcal{L}^{(l)}$$

Hyperparameter Definition

- ▶ $\mathcal{L}_{CE}$: Classical cross-entropy loss driving target category discrimination.
- ▶ α : Scaling weight setting the optimization magnitude of the contrastive regularization branch.

The Bottleneck

Pairwise vector products across multiple hidden layers $\implies$ Severe memory bottlenecks.

The Matrix Parallelization Solution

- ▶ **Instant Computation:** Stack batch vectors into matrix $U^{(l)} \rightarrow U^{(l)T}U^{(l)}$.
- ▶ **Minimal Overhead:** Denominator computed **once per batch** (identical for all classes).

Optimizer Configurations

- ▶ Base Optimization: Stochastic Gradient Descent (SGD) for 30 epochs.
- ▶ Learning Rate: 0.001, Batch Size Allocation: 128.
- ▶ Default Extraction Blocks Settings: Reduction factor $r = 4$.
- ▶ Default Loss Balancing: Weight $\alpha = 0.01$, Temperature $\tau = 1.0$.



VLCS Dataset

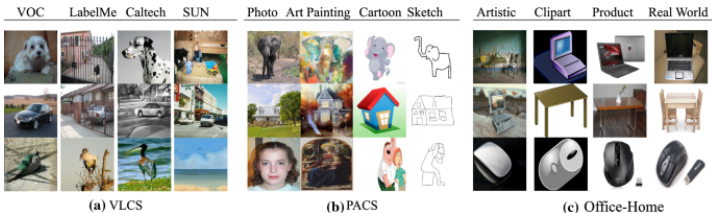
- **Size:** 10,729 images —
Classes: 5
- **Domains (4):** PASCAL, LabelMe, Caltech, SUN.

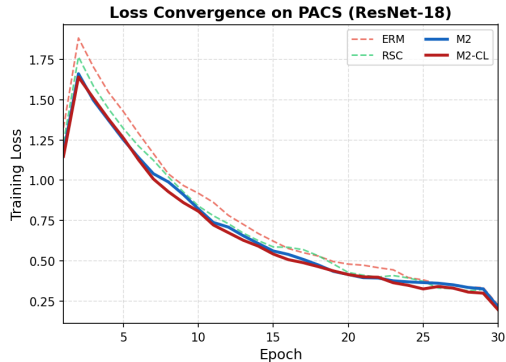
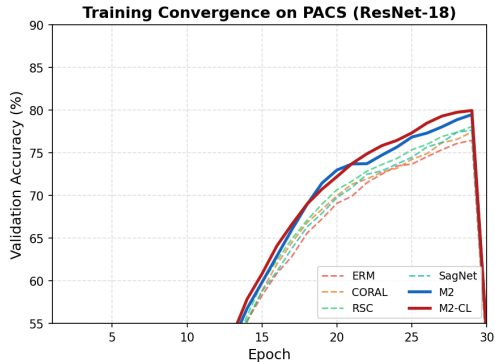
PACS Dataset

- **Size:** 9,991 images —
Classes: 7
- **Domains (4):** Photo, Art Painting, Cartoon, Sketch.

Office-Home Dataset

- **Size:** 15,588 images —
Classes: 65
- **Domains (4):** Art, Clipart, Product, Real-World.





M^2 -CL converges faster and plateaus **higher** than all baselines. The contrastive regularizer suppresses loss spikes caused by domain-conflicting gradients.

Top-1 accuracy on PACS and VLCS, ResNet-18.

Method	Art	Cartoon	Photo	Sketch	Avg	Caltech	Labelme	Sun	Voc	Avg
ERM	76.53	74.80	94.95	67.89	78.54	96.87	65.03	71.62	71.60	76.28
RSC	79.26	76.12	95.07	70.88	80.34	97.68	66.33	73.53	72.75	77.57
CORAL	78.55	72.74	92.24	73.93	79.36	95.16	60.73	70.66	65.00	72.89
MIXUP	79.79	72.81	93.11	63.97	77.42	93.44	60.25	74.36	69.87	74.48
MMD	77.75	73.35	92.34	69.74	78.29	97.28	60.62	70.48	65.21	73.40
SagNet	78.71	74.90	94.79	70.48	79.72	95.76	61.22	69.61	71.90	74.62
SelfReg	79.46	76.32	95.93	71.60	80.83	92.53	62.56	70.27	72.15	74.38
ARM	79.77	75.33	94.43	<u>74.76</u>	81.07	95.86	60.19	70.66	72.83	74.89
EQRM	78.61	72.38	92.48	71.13	78.65	96.17	60.57	70.44	64.66	72.96
SAGM	78.22	75.14	94.79	70.46	79.65	96.37	66.22	71.14	71.35	76.27
M2	<u>80.64</u>	<u>76.91</u>	95.50	74.20	<u>81.81</u>	<u>98.24</u>	<u>67.02</u>	73.80	<u>72.92</u>	<u>78.00</u>
M2-CL	81.04	77.48	<u>95.70</u>	75.81	82.51	98.69	67.48	<u>74.05</u>	73.76	78.50

Top-1 accuracy on PACS and VLCS, ResNet-50.

Method	Art	Cartoon	Photo	Sketch	Avg	Caltech	Labelme	Sun	Voc	Avg
ERM	83.54	78.47	96.51	73.64	83.04	97.01	62.22	71.79	76.09	76.78
RSC	82.08	80.26	96.51	75.45	83.57	97.81	64.44	74.60	75.53	78.10
CORAL	83.94	79.27	94.37	81.98	84.89	97.01	61.34	75.26	71.74	76.34
MIXUP	85.16	77.40	96.23	78.26	84.26	96.94	62.33	73.40	72.32	76.25
MMD	82.21	74.70	92.89	69.00	79.70	97.48	63.85	75.49	71.65	77.12
SagNet	85.23	78.47	93.20	77.71	83.65	94.14	60.43	70.49	69.96	73.75
SelfReg	83.33	79.06	93.16	77.24	83.20	92.49	60.73	66.80	73.58	73.40
ARM	82.14	79.10	92.97	73.02	81.81	96.01	63.98	66.26	71.79	74.51
EQRM	84.98	77.74	91.26	74.88	82.21	94.35	60.80	66.35	72.94	73.61
SAGM	81.35	78.99	93.24	79.39	83.24	97.39	64.23	69.30	76.17	76.77
M2	<u>86.02</u>	79.90	<u>97.03</u>	81.50	<u>86.11</u>	97.52	<u>64.77</u>	<u>76.02</u>	<u>76.72</u>	<u>78.76</u>
M2-CL	86.48	<u>80.05</u>	97.99	<u>81.70</u>	86.56	<u>97.60</u>	64.86	76.65	77.26	79.09

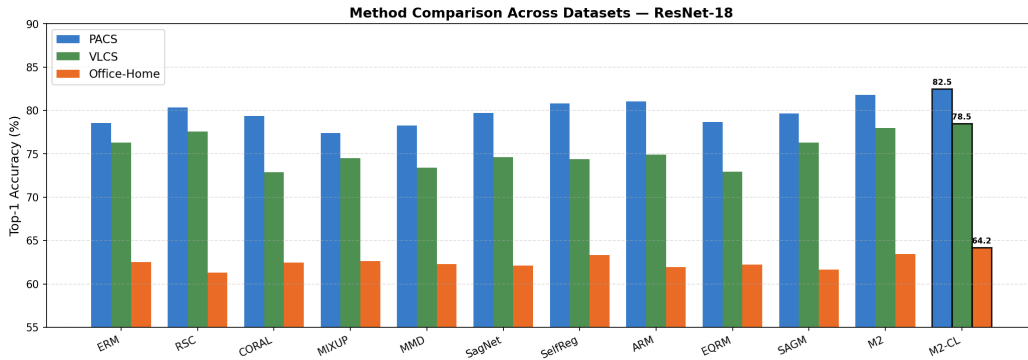
- ▶ **State-of-the-Art:** M^2 -CL sets a new state-of-the-art benchmark average across both backbones.
- ▶ **The Contrastive Loss Boost:** M^2 -CL consistently outperforms the plain M^2 architecture, confirming that its explicit geometric alignment helps.
- ▶ **The Sketch Challenge Exception:** Performance drops slightly in the Sketch target domain[cite: 362]. This is likely because the dominant white backgrounds make it difficult for the network to separate structural objects from the domain style[cite: 363].

Top-1 accuracy on Office-Home, ResNet-18.

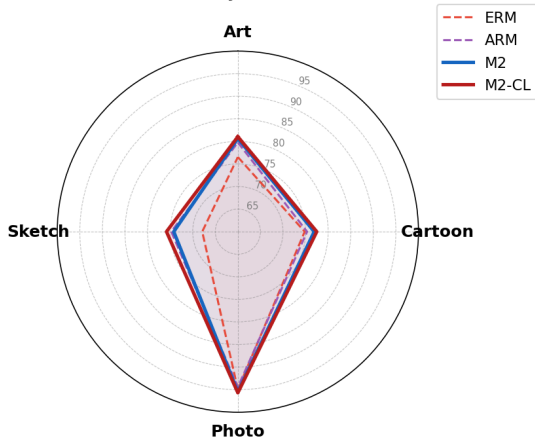
Method	Art	Clipart	Product	Real-World	Avg
ERM	56.12	47.88	<u>72.47</u>	73.58	62.51
RSC	56.00	46.09	<u>70.85</u>	72.32	61.31
CORAL	55.46	49.39	71.75	73.28	62.47
MIXUP	56.37	<u>49.81</u>	70.87	73.61	62.66
MMD	55.67	48.52	71.68	73.19	62.27
SagNet	55.75	47.74	71.71	73.17	62.09
SelfReg	57.56	49.69	72.43	73.72	63.35
ARM	54.47	48.45	71.14	<u>73.79</u>	61.96
EQRM	56.08	48.34	71.57	72.94	62.23
SAGM	55.79	46.51	71.16	73.15	61.65
M2	<u>58.04</u>	49.30	73.04	73.50	<u>63.47</u>
M2-CL	58.77	50.86	72.20	74.91	64.18

Top-1 accuracy on Office-Home, ResNet-50.

Method	Art	Clipart	Product	Real-World	Avg
ERM	63.43	51.32	76.45	<u>78.13</u>	67.33
RSC	62.64	51.55	75.11	77.34	66.66
CORAL	61.27	<u>54.34</u>	73.35	77.83	66.70
MIXUP	63.23	51.91	<u>77.10</u>	76.39	67.16
MMD	59.63	49.37	75.28	74.71	64.75
SagNet	61.41	51.85	72.85	73.79	64.97
SelfReg	59.95	51.22	71.28	74.52	64.24
ARM	54.23	52.14	71.73	71.26	62.34
EQRN	63.52	52.32	73.28	76.45	66.39
SAGM	61.01	53.27	74.81	77.00	66.52
M2	<u>65.35</u>	53.90	76.80	78.74	<u>68.70</u>
M2-CL	66.46	54.82	78.24	77.85	69.34



Per-Domain Accuracy on PACS (ResNet-18)



Key Observations

- ▶ M^2 -CL dominates **all four** domains simultaneously.
- ▶ Largest gain: **Sketch** (+7.9 pp over ERM), the hardest domain due to extreme style shift.
- ▶ **Art Painting** gap confirms that contrastive regularization helps most when style variance is high.

Why ARM fails on Photo

ARM's meta-learning relies on test-time adaptation signals; **Photo** (near-real domain) provides weak adaptation gradient,

Average Accuracy

- ▶ **ResNet-18:** M^2 -CL reaches **82.51** on PACS, **78.50** on VLCS, and **64.18** on Office-Home.
- ▶ **ResNet-50:** M^2 -CL reaches **86.56** on PACS, **79.09** on VLCS, and **69.34** on Office-Home.
- ▶ The average score improves over the strongest baseline in each benchmark summary.

Interpretation

- ▶ M^2 improves representation quality by reusing intermediate multi-scale features.
- ▶ M^2 -CL adds class-level alignment, making same-class features closer across source domains.
- ▶ Per-domain scores are not uniformly best, so the ablation section checks which design choices drive the average gains.

Ablation study on PACS and VLCS, ResNet-18.

pipe	r	drop	loss	A	C	P	S	PACS Avg	C	L	S	V	VLCS Avg
c	2	-	-	72.34	72.29	94.19	69.51	78.23	93.87	60.91	65.89	61.91	71.22
c	4	-	-	74.38	72.74	90.13	69.71	77.21	94.14	59.01	67.82	65.42	69.62
c	6	-	-	75.72	70.12	94.26	73.33	77.92	91.93	56.81	67.28	64.10	68.95
c	2	yes	-	75.67	73.29	92.16	75.95	77.93	93.73	57.72	66.34	62.50	71.74
c	4	yes	-	74.36	74.05	92.57	76.45	79.37	96.07	56.28	67.07	64.30	65.81
c	6	yes	-	75.97	72.77	93.01	74.73	76.81	92.94	59.73	66.15	64.51	70.09
p	2	-	-	76.09	55.88	68.35	60.92	64.40	95.60	56.97	66.25	73.15	73.96
p	4	-	-	74.72	74.62	92.01	70.32	77.64	95.20	58.10	66.20	64.90	70.29
p	6	-	-	75.38	69.99	91.21	69.47	77.77	95.23	56.81	69.13	65.34	70.55
p	2	yes	-	74.61	72.39	87.08	73.95	78.88	95.73	58.33	68.44	63.74	70.49
p	6	yes	-	76.73	73.98	93.41	73.56	79.84	93.19	56.05	66.62	64.24	70.43
p	4	yes	-	76.39	75.98	95.58	74.49	79.15	94.54	59.77	67.26	74.55	73.14
p	4	yes	yes	80.43	75.43	96.87	74.42	80.10	96.09	61.15	69.05	75.70	74.15

Sensitivity analysis for τ in M2-CL, ResNet-18.

τ	PACS	VLCS
0.01	78.10	76.15
0.1	80.03	77.27
0.2	81.37	74.71
0.4	82.07	72.03
0.6	83.70	74.15
0.8	84.67	75.21
1.0	80.07	76.24
1.2	81.34	75.11
1.4	83.14	75.69
1.6	82.17	76.84
1.8	80.51	74.66
2.0	78.72	74.99
10.0	79.92	74.04
100.0	81.84	73.74



Sensitivity analysis for α in M2-CL, ResNet-18.

α	PACS	VLCS
0.00	82.76	72.36
10e-5	80.41	73.56
10e-4	81.17	73.71
10e-3	82.29	74.38
10e-2	79.70	73.39
10e-1	64.04	55.96

Table 5 Reference Insights (PACS / VLCS, ResNet-18)

- ▶ **Best full configuration:** $p, r = 4, drop = yes, loss = yes$ reaches **80.10** PACS Avg and **74.15** VLCS Avg.
- ▶ **Contrastive loss contribution:** adding the loss to the same $p, r = 4, drop = yes$ setting improves PACS from **79.15** to **80.10** and VLCS from **73.14** to **74.15**.
- ▶ **Spatial dropout effect:** for $p, r = 4$, dropout improves PACS from **77.64** to **79.15** and VLCS from **70.29** to **73.14**.

Tables 6–7 Insights: Hyperparameter Tuning

- ▶ **Temperature optimum:** PACS peaks at **84.67** with $\tau = 0.8$, while VLCS peaks at **77.27** with $\tau = 0.1$.
- ▶ **Default-scale behavior:** $\tau = 1.0$ remains competitive with **80.07** PACS and **76.24** VLCS.
- ▶ **Loss-weight sensitivity:** $\alpha = 10e^{-3}$ gives the best VLCS score, **74.38**, while $\alpha = 0$ gives the highest PACS score, **82.76**.

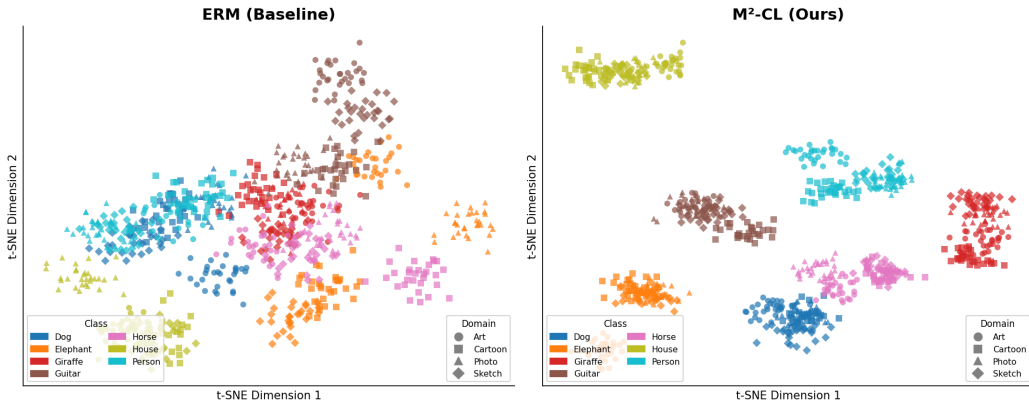
Alpha Bounds Limits

$$\alpha = 10e^{-1} \implies \text{PACS } 64.04, \quad \text{VLCS } 55.96$$

A large contrastive weight dominates cross-entropy optimization and causes a sharp drop compared with the best scores in Table 7.



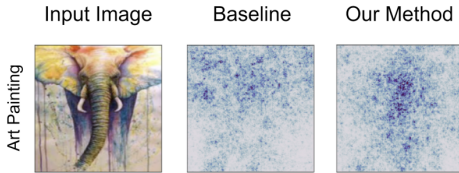
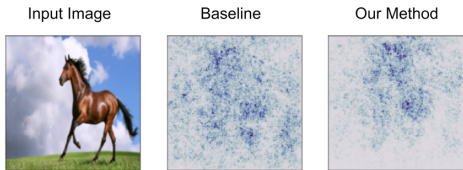
Feature Space Visualization — PACS (Sketch target domain, ResNet-18)



Left (ERM): Same-class features from different domains are dispersed — the model learns domain-specific patterns. **Right (M^2 -CL):** Contrastive regularization collapses domain scatter within each class, while maintaining inter-class structure.

Methodology

Visualizes the gradients of the class score function relative to input image pixels. Darker pixel weights highlight the regions that contribute most to final classification decisions.



Saliency maps comparison on the NICO and PACS datasets.

Baseline ERM Traps

- ▶ Strongly influenced by background noise and spurious environmental features (e.g., sky, roads, grass).
- ▶ Often focuses on non-causal elements within the scene, such as a motorcycle rider.

Proposed M^2 -CL Solution

- ▶ Successfully ignores irrelevant background styles and focus fields.
- ▶ Accurately isolates the core structural contours of objects, even in highly stylized domains like Sketch and Cartoon.

Core Achievements

- ▶ **Architecture:** Successfully combines multi-scale, multi-layer intermediate features without propagating domain-specific styles through the network.
- ▶ **Loss Optimization:** Introduces a layer-wise contrastive loss that enforces domain invariance across all extraction levels.
- ▶ **Empirical Validation:** Outperforms previous domain generalization methods and sets a new state-of-the-art across four major benchmark datasets.

Current Computational Drawbacks

- ▶ **Memory Overhead:** Concatenating multiple intermediate feature maps creates dense hypercolumns before the classification head, increasing memory usage.
- ▶ **Batch Size Dependency:** Like most contrastive methods, M^2 -CL requires large training batches to construct effective positive/negative pairs.

Future Research Avenues

Explore cross-layer attention mechanics, build causal inference systems, and evaluate alternative similarity metrics like KL divergence to refine latent distribution shapes.



A decorative graphic on the left side of the slide. It features a dark blue background with a large, stylized circular pattern composed of many small red dots. The dots are arranged in concentric, slightly offset rings, creating a sense of depth and movement. The word "HUST" is centered within this pattern.

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